

Seminars 11 & 12: Computing the Spectrum

Questions and Final Answers

1. From the Fourier integral to the DFT

A continuous-time signal is sampled with sampling period T_s , producing $x[n] = x_c(nT_s)$, $n = 0, \dots, N-1$. The continuous-time Fourier transform is $X_c(j\omega) = \int_{-\infty}^{\infty} x_c(t) e^{-j\omega t} dt$

(a) Approximate the Fourier integral using a finite Riemann sum based on the N available samples.

Answer.

$$X_c(j\omega) \approx T_s \sum_{n=0}^{N-1} x[n] e^{-j\omega n T_s}$$

(b) Choose the analysis frequencies $\omega_k = \frac{2\pi k}{NT_s}$, $k = 0, \dots, N-1$

Answer.

$$\omega_k = \frac{2\pi k}{NT_s}, \quad k = 0, \dots, N-1$$

(c) and prove that the resulting expression is proportional to the DFT $X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}$

Answer.

$$X_c(j\omega_k) \approx T_s X[k]$$

(d) Explain the role of T_s in converting the DFT coefficients into an approximation of the CTFT.

Answer. The DFT is dimensionless with respect to the sample index. Multiplication by T_s converts the discrete sum into an approximation of integration over physical time.

(e) Identify three distinct sources of discrepancy between an FFT-computed spectrum and the ideal CTFT.

Answer. The three effects are conceptually distinct: sampling in time can alias the continuous-time spectrum; keeping only N samples multiplies the signal by a finite window and causes convolution/leakage in frequency; and the DFT evaluates the resulting spectrum only at N discrete frequencies.

2. The four Fourier representations

Complete a comparison of the continuous-time Fourier transform, discrete-time Fourier transform, Fourier series, and discrete Fourier transform.

(a) For each representation, state whether time is continuous or discrete.

Answer. CTFT and Fourier series describe continuous-time signals in this course. DTFT and DFT start from sequences $x[n]$.

(b) State whether frequency is continuous or discrete.

Answer. The CTFT uses continuous ω ; the DTFT uses continuous normalized frequency ω . Fourier-series coefficients are indexed by integer harmonics, and the DFT is indexed by integer bins k .

(c) State whether the frequency representation is periodic.

Answer. DTFT periodicity follows from $e^{-j(\hat{\omega}+2\pi)n} = e^{-j\hat{\omega}n}$. DFT coefficients repeat when k is increased by N . The other two representations do not in general have periodic coefficient functions.

(d) State whether the signal is assumed finite, periodic, or potentially infinite.

Answer. Each transform is associated with a different signal model. The DFT treats the N supplied samples as one period of a periodic discrete-time sequence.

(e) Write the corresponding analysis equation.

Answer.

$$\text{CTFT: } X_c(j\omega) = \int_{-\infty}^{\infty} x_c(t) e^{-j\omega t} dt$$

$$\text{DTFT: } X(e^{j\hat{\omega}}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\hat{\omega}n}$$

$$\text{FS: } c_k = \frac{1}{T_0} \int_{T_0} x(t) e^{-jk\omega_0 t} dt$$

$$\text{DFT: } X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}$$

(f) Explain why all four representations can be interpreted as projections of a signal onto complex exponentials.

Answer. In each case the analysis operation correlates or projects the signal onto $e^{j\omega t}$ or $e^{j\hat{\omega}n}$. The main differences are whether time/frequency are continuous or discrete and whether the signal is finite or periodic.

3. Sampling, DTFT periodicity, and aliasing

The continuous-time signal $x_c(t) = 2\cos(2\pi 600t) + \cos(2\pi 1300t)$

is sampled at $f_s = 2000$ samples/s

(a) Determine the normalized radian frequencies of both sinusoidal components.

Answer.

$$\hat{\omega}_{600} = 0.6\pi$$

$$\hat{\omega}_{1300} = 1.3\pi \equiv -0.7\pi$$

(b) Determine whether either component violates the sampling theorem.

Answer. The sampling theorem requires every analogue component to lie below $\frac{f_s}{2} = 1000$ Hz. Therefore the 1300-Hz component aliases.

(c) Find the aliases inside $0 \leq f \leq \frac{f_s}{2}$.

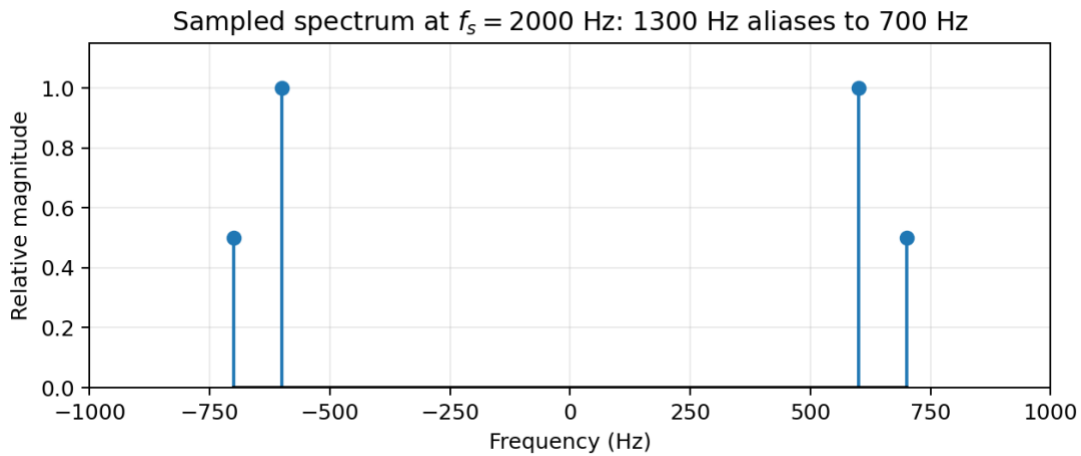
Answer.

$$f_{\text{alias}} = |1300 - 2000| = 700 \text{ Hz}$$

Thus the sampled real signal contains 600-Hz and 700-Hz components.

(d) Sketch one period of the magnitude DTFT of the sampled signal.

Answer.



Aliased line spectrum for $f_s = 2000$ Hz.

(e) Predict the DFT-bin locations of the peaks for $N = 100$.

Answer.

Hence the four line-spectrum peaks are at bins 30, 35, 65, and 70.

(f) Repeat the analysis for $f_s = 4000$ samples/s and explain the difference.

Answer.

$$\hat{\omega}_{600} = 2\pi(600/4000) = 0.3\pi$$

$$\hat{\omega}_{1300} = 2\pi(1300/4000) = 0.65\pi$$

$$\Delta f = 4000/100 = 40 \text{ Hz}$$

600 Hz is exactly bin 15. The 1300-Hz component corresponds to bin 32.5, so it lies halfway between bins 32 and 33 and leaks into neighbouring bins.

4. Direct calculation of a short DFT and IDFT

Consider the four-sample sequence $x[n] = \{1, 1, 0, 0\}$, $0 \leq n \leq 3$

(a) Calculate its four-point DFT directly from the definition.

Answer.

$$X[0] = 2$$

$$X[1] = 1 - j$$

$$X[2] = 0$$

$$X[3] = 1 + j$$

(b) Express every coefficient in rectangular and polar form.

Answer.

$$X[0] = 2 = 2\angle 0$$

$$X[1] = 1 - j, |X[1]| = \sqrt{2}, \angle X[1] = -\pi/4$$

$$X[2] = 0$$

$$X[3] = 1 + j, |X[3]| = \sqrt{2}, \angle X[3] = \pi/4$$

(c) Verify conjugate symmetry.

Answer.

$$X[4 - 1] = X[3] = 1 + j = (1 - j)^* = X^*[1]$$

Bins 0 and 2 are self-conjugate and are therefore real.

(d) Use the inverse DFT to reconstruct $x[n]$.

Answer.

$$x[n] = \frac{1}{4} \sum_{k=0}^3 X[k] e^{j2\pi kn/4} = \{1, 1, 0, 0\}$$

(e) Identify the physical frequency represented by each bin when $f_s = 8000$ samples/s.

Answer.

$$k = 0 \rightarrow 0 \text{ Hz}$$

$$k = 1 \rightarrow 2000 \text{ Hz}$$

$$k = 2 \rightarrow 4000 \text{ Hz (Nyquist)}$$

$$k = 3 \rightarrow 6000 \text{ Hz} \equiv -2000 \text{ Hz}$$

(f) Explain why bins $k = 3$ and $k = -1$ represent the same discrete-time frequency.

Answer. The DFT basis is periodic in k with period N , so $e^{j2\pi(3)n/4} = e^{j2\pi(-1)n/4}$.

5. DFT bins and negative frequencies

A real signal is sampled at 16000 samples/s and analysed with a 16-point DFT.

(a) Calculate the frequency spacing Δf .

Answer.

$$\Delta f = f_s/N = 16000/16 = 1000 \text{ Hz}$$

(b) List the physical frequency represented by every bin $k = 0, \dots, 15$, using negative-frequency labels for the upper half.

Answer.

$$k: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15$$

$$f_k \text{ (kHz): } 0, 1, 2, 3, 4, 5, 6, 7, 8, -7, -6, -5, -4, -3, -2, -1$$

(c) Identify the bins representing DC, the Nyquist frequency, +3000 Hz, and -3000 Hz.

Answer.

$$DC \rightarrow k = 0$$

$$\text{Nyquist } 8000 \text{ Hz} \rightarrow k = 8$$

$$+3000 \text{ Hz} \rightarrow k = 3$$

$$-3000 \text{ Hz} \rightarrow k = 16 - 3 = 13$$

(d) If $X[3] = 4 - 2j$, determine $X[13]$.

Answer. For a real signal $X[N - k] = X^*[k]$. Since $13 = 16 - 3$, $X[13] = X^*[3] = 4 + 2j$.

(e) State which DFT coefficients are necessarily real for an even-length DFT of a real signal.

Answer. DC and the Nyquist bin are their own conjugate partners for $N=16$, so their imaginary parts must vanish.

(f) Explain why plotting all bins from 0 to f_s without relabelling the upper half can be misleading.

Answer. Without relabelling, the second half of the DFT can be mistaken for additional high positive-frequency content rather than the conjugate negative-frequency half of a real signal.

6. Bin-centred and non-bin-centred sinusoids

Let $x[n] = \cos\left(\frac{2\pi f_0 n}{f_s}\right)$

with $f_s = 1024$ samples/s and $N = 256$. Analyse the cases $f_0 = 128$ Hz and $f_0 = 130$ Hz.

(a) Calculate the DFT-bin spacing.

Answer.

$$\Delta f = 1024/256 = 4 \text{ Hz}$$

(b) Determine whether each sinusoid is bin-centred.

Answer.

$$128/4 = 32 \rightarrow \text{integer} \rightarrow \text{bin-centred}$$

$$130/4 = 32.5 \rightarrow \text{non-integer} \rightarrow \text{not bin-centred}$$

(c) Predict the positive- and negative-frequency peak locations.

Answer.

$$128 \text{ Hz} \rightarrow k = 32 \text{ and } k = 224$$

For 130 Hz, the ideal line is centred halfway between bins 32 and 33; the negative-frequency line is correspondingly halfway between bins 223 and 224. Energy therefore spreads over neighbouring DFT bins.

(d) Explain why one case ideally produces energy in only two bins while the other spreads energy over many coefficients.

Answer. For 128 Hz there are exactly 32 cycles in 256 samples. The record endpoints join coherently under the DFT periodic-extension assumption. The 130-Hz record has 32.5 cycles, producing a boundary discontinuity and leakage.

(e) Describe the relationship between the non-bin-centred DFT samples and a shifted Dirichlet-function envelope.

Answer. A finite rectangularly windowed complex exponential has a Dirichlet-kernel spectrum. When the sinusoid is off-bin, the DFT samples this continuous envelope away from its peak.

(f) Explain why this spreading is not an FFT error.

Answer. FFT and direct DFT give the same coefficients. Leakage belongs to the signal/window model, not to numerical failure of the FFT algorithm.

7. Rectangular-window spectrum and spectral leakage

A sinusoid is observed only over the interval $0 \leq t < T$, which is equivalent to multiplying it by a rectangular window.

(a) Derive the CTFT of the duration- T rectangular window.

Answer.

$$W(j\omega) = \int_0^T e^{-j\omega t} dt = T e^{-j\omega T/2} \text{sinc}\left(\frac{\omega T}{2}\right)$$

(b) Determine the locations of the first zeros.

Answer.

$$\text{first zeros: } \omega = \pm 2\pi/T$$

(c) Show that the mainlobe width between the first zeros is $\Delta\omega = \frac{4\pi}{T}$

Answer.

$$\Delta\omega = \frac{4\pi}{T}$$

(d) Describe how the ideal spectrum of an infinite-duration sinusoid changes after windowing.

Answer. Multiplication by the window in time becomes convolution in frequency. Convolution of an ideal delta line with $W(j\omega)$ produces a shifted sinc/Dirichlet-shaped lobe.

(e) Explain the origin of spectral leakage.

Answer. The rectangular window has slowly decaying sidelobes. Those sidelobes spread the energy of a single sinusoid over neighbouring frequencies.

(f) Determine how the mainlobe width changes if the observation duration is doubled.

Answer.

$$\Delta\omega_{new} = 4\pi/(2T) = 2\pi/T = \Delta\omega/2$$

Doubling the observation duration halves the mainlobe width.

(g) Explain why sidelobes may be mistaken for weak sinusoidal components.

Answer. A strong component can create sidelobe levels larger than a nearby weak component, so visual peak detection can be misleading.

8. Rectangular versus Hamming windows

Two equal-amplitude sinusoids have frequencies 1000 Hz and 1080 Hz and are sampled at $f_s = 8000$ samples/s. Use the approximate first-zero mainlobe widths $\Delta f_{rect} \approx \frac{2}{T}$ $\Delta f_{Hamming} \approx \frac{4}{T}$

(a) Estimate the minimum observation duration required to resolve the two sinusoids with a rectangular window.

Answer.

$$T \geq 2/80 = 0.025 \text{ s} = 25 \text{ ms}$$

(b) Repeat for a Hamming window.

Answer.

$$T \geq 4/80 = 0.050 \text{ s} = 50 \text{ ms}$$

(c) Explain why the Hamming window needs a longer duration for comparable frequency resolution.

Answer. The Hamming window trades resolution for much lower sidelobes. Therefore a longer record is needed to recover the same mainlobe width.

(d) State which window is preferable when detecting a weak sinusoid close to a strong sinusoid.

Answer. When a weak sinusoid lies near a strong one, low sidelobes reduce leakage from the strong component and improve dynamic-range discrimination.

(e) State which is preferable when separating two nearby components of comparable amplitude.

Answer. Its narrower mainlobe gives better resolution for a fixed T, provided sidelobe masking is not the limiting issue.

(f) Discuss the trade-off between mainlobe width and sidelobe level.

Answer. This is the central window-design trade-off: rectangular gives a narrow mainlobe but high sidelobes; Hamming gives a wider mainlobe but much lower sidelobes.

9. Resolving a windowed sum of sinusoids

Consider $x_c(t) = 1 + 0.8\cos(2\pi 200t) + 0.15\cos(2\pi 230t)$ sampled at $f_s = 2000$ samples/s.

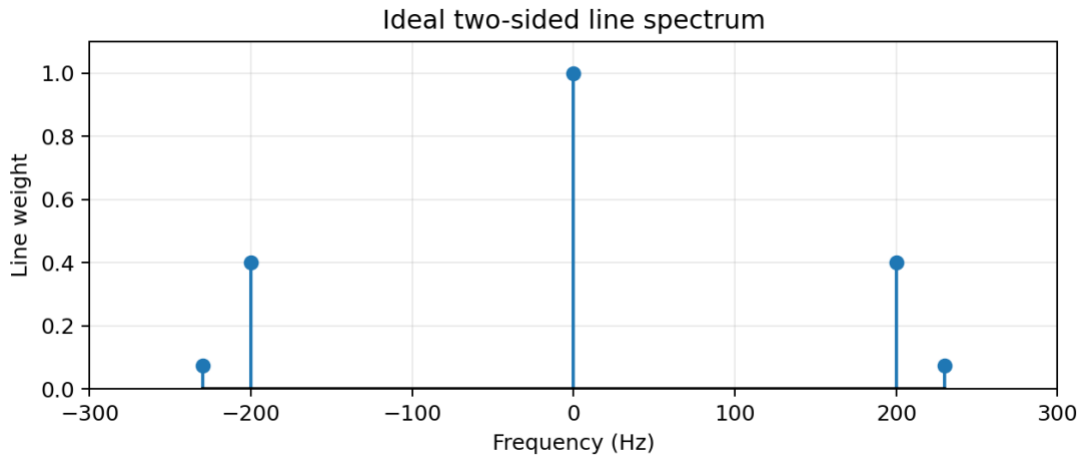
(a) Draw the ideal two-sided CTFT line spectrum.

Answer.

DC: $f = 0$, weight 1

200 – Hz cosine: weights $0.8/2 = 0.4$ at ± 200 Hz

230 – Hz cosine: weights $0.15/2 = 0.075$ at ± 230 Hz



Ideal two-sided line spectrum.

(b) Predict the spectrum after observation with a 20-ms rectangular window.

Answer.

$$T = 0.020 \text{ s} \rightarrow \Delta f_{\text{rect}} \approx 2/T = 100 \text{ Hz}$$

The two positive-frequency lines are only 30 Hz apart, so their 100-Hz-wide mainlobes overlap strongly and are not resolved.

(c) Repeat for a 100-ms rectangular window.

Answer.

$$T = 0.100 \text{ s} \rightarrow \Delta f_{\text{rect}} \approx 2/T = 20 \text{ Hz}$$

Now the 30-Hz separation exceeds the approximate 20-Hz mainlobe width, so the two peaks can be resolved much more clearly.

(d) Determine in which case the 200-Hz and 230-Hz components are likely to be resolved.

Answer.

$$20 \text{ ms: } 30 \text{ Hz} < 100 \text{ Hz} \rightarrow \text{unresolved}$$

$$100 \text{ ms: } 30 \text{ Hz} > 20 \text{ Hz} \rightarrow \text{likely resolved}$$

(e) Discuss whether the weak 230-Hz component may be obscured by leakage from the 200-Hz component.

Answer. The weak component is only 18.75% of the strong component in amplitude. Rectangular-window sidelobes from the strong line can therefore mask it.

(f) Explain how a Hamming window changes the result.

Answer. At 100 ms the approximate Hamming first-zero width is 40 Hz, so the 30-Hz-separated peaks may still overlap even though leakage is much lower. This illustrates the resolution-versus-dynamic-range trade-off.

(g) Distinguish failure to resolve components from insufficient DFT-bin density.

Answer. A smooth zero-padded plot can still contain unresolved components. Resolution is controlled primarily by the amount of actual data and the window, not merely by the FFT length.

10. Zero-padding and true frequency resolution

A length- $L = 40$ sinusoidal sequence is analysed using: (a) a 40-point DFT; (b) a 256-point DFT after appending zeros; and (c) a 256-point DFT of 256 actual signal samples.

(a) Compare the DFT-bin spacings.

Answer.

$$\begin{aligned} 40 - \text{point DFT: } \Delta f &= f_s/40 \\ 40 \text{ samples} + \text{zero-padding to 256: } \Delta f &= f_s/256 \\ 256 \text{ actual samples, 256-point DFT: } \Delta f &= f_s/256 \end{aligned}$$

The last two have the same grid spacing, but not the same physical resolution.

(b) Identify which two analyses have the same window mainlobe width.

Answer. The first and second analyses both observe only 40 actual samples, so both use the same 40-sample rectangular data window.

(c) Identify which analysis has the best true ability to resolve two nearby sinusoids.

Answer.

$$256/40 = 6.4$$

Its observation duration is 6.4 times longer, so its rectangular-window mainlobe is 6.4 times narrower and its true resolving ability is best.

(d) Explain why zero-padding produces a smoother-looking spectrum without creating new information.

Answer. The 256-point zero-padded DFT interpolates between the original 40-point frequency samples of the same windowed signal spectrum.

(e) State one practical benefit of zero-padding.

Answer. Denser frequency sampling is useful for plotting and for interpolating the location of an isolated spectral maximum.

(f) State one incorrect conclusion that could be drawn from the smoother plot.

Answer. The smoother curve can look sharper simply because more frequency points are shown; the window mainlobe has not become narrower.

11. Spectrum of an FIR filter

The impulse response of an FIR filter is $h[n] = \{1, 2, 3, 2, 1\}$

(a) Compute its five-point DFT.

Answer.

$$X = [9, -2.118 - 1.539j, 0.118 + 0.363j, 0.118 - 0.363j, -2.118 + 1.539j]$$

(b) Compute a 64-point DFT after zero-padding the impulse response.

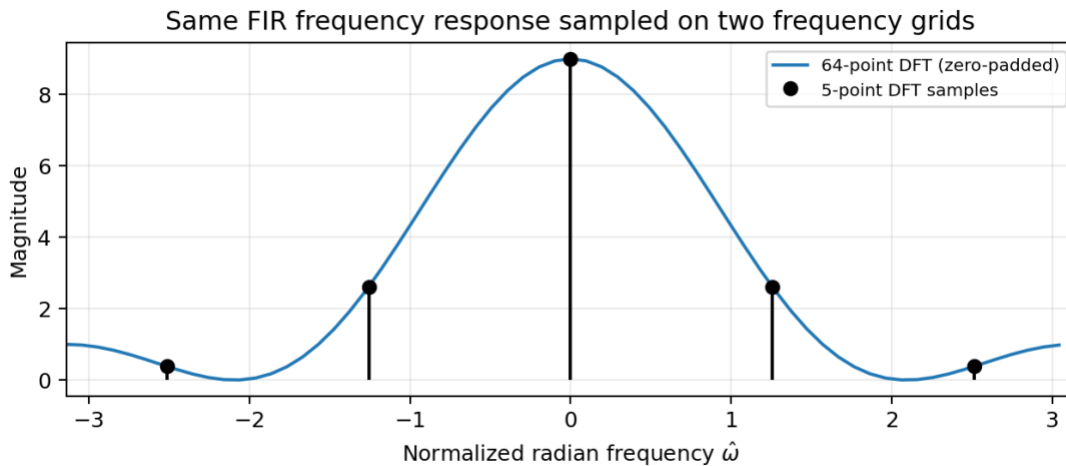
Answer.

$$X_{64}[k] = \sum_{n=0}^4 h[n] e^{-j2\pi kn/64}, \quad k = 0, \dots, 63$$

This is the same continuous FIR frequency response evaluated at 64 equally spaced unit-circle frequencies; zero-padding changes the sampling grid, not $h[n]$.

(c) Plot both magnitude responses against normalized radian frequency.

Answer.



Five-point and zero-padded 64-point DFT magnitude responses.

(d) Explain why the 64-point result appears smoother.

Answer. The underlying response is continuous in $\hat{\omega}$; zero-padding simply samples it more densely.

(e) State whether the filter itself has changed.

Answer. Its impulse response and coefficients remain $\{1, 2, 3, 2, 1\}$. Only the frequency grid used to evaluate the response changes.

(f) Determine the DC gain directly from $h[n]$ and verify it in the DFT.

Answer.

$$H(e^{j0}) = 1 + 2 + 3 + 2 + 1 = 9$$

(g) Explain how increasing the actual impulse-response length can sharpen filter features, whereas increasing only the DFT length cannot.

Answer. Filter selectivity is determined by the coefficients and their length. FFT length is an analysis parameter, not a filter-design parameter.

12. Periodic signals and coherent sampling

Consider the periodic discrete-time signal $x[n] = 2 + \cos\left(\frac{\pi n}{4}\right) + 0.5\cos\left(\frac{\pi n}{2} + \frac{\pi}{3}\right)$

(a) Determine the fundamental period.

Answer.

$$N_0 = \text{lcm}(8, 4) = 8$$

(b) Compute the DFT of exactly one period.

Answer.

$$X[0] = 16$$

$$\text{first cosine: } X[1] = 4, \quad X[7] = 4$$

$$\text{second cosine: } X[2] = 2e^{j\pi/3} = 1 + j\sqrt{3}$$

$$X[6] = 2e^{-j\pi/3} = 1 - j\sqrt{3}$$

$$X[3] = X[4] = X[5] = 0$$

(c) Identify the bins containing nonzero coefficients.

Answer.

$$\text{nonzero bins: } k = 0, 1, 2, 6, 7$$

These correspond to DC, $\pm\pi/4$, and $\pm\pi/2$.

(d) Relate the DFT coefficients to the Fourier-series coefficients.

Answer.

$$c_k = X[k]/8$$

(e) Use the inverse DFT to reconstruct one period.

Answer.

$$x[n] = 2 + \cos(\pi n/4) + 0.5 \cos(\pi n/2 + \pi/3)$$

(f) Predict what happens if a record containing 1.5 periods is transformed.

Answer. The DFT implicitly repeats the 12-sample record. Since the repeated record does not join at the same phase, the artificial boundary creates spectral spreading.

(g) Explain why selecting an integer number of periods avoids artificial boundary discontinuities and leakage.

Answer. Coherent sampling makes the periodic extension assumed by the DFT agree with the actual periodic signal.

14. Constructing and calibrating a spectrogram

A signal is sampled at $f_s = 16000$ samples/s. A spectrogram is calculated using window length $L = 400$, FFT length $N = 1024$, and hop size $R = 160$.

(a) Determine the window duration in milliseconds.

Answer.

$$T_w = L/f_s = 400/16000 = 0.025 \text{ s} = 25 \text{ ms}$$

(b) Determine the time separation between consecutive frames.

Answer.

$$T_{hop} = R/f_s = 160/16000 = 0.010 \text{ s} = 10 \text{ ms}$$

(c) Calculate the percentage overlap.

Answer.

$$\text{overlap} = 240/400 \times 100\% = 60\%$$

(d) Determine the DFT-bin spacing in hertz.

Answer.

$$\Delta f_{bin} = f_s/N = 16000/1024 = 15.625 \text{ Hz}$$

(e) Using the approximation $\Delta f_{\text{Hamming}} \approx \frac{4f_s}{L}$

(f) estimate the effective frequency resolution.

Answer.

$$\Delta f_{\text{Hamming}} \approx 4f_s/L = 4 \cdot 16000/400 = 160 \text{ Hz}$$

(g) Explain why DFT-bin spacing and effective frequency resolution are not the same.

Answer. Zero-padding can reduce bin spacing without changing the mainlobe width. Therefore a very dense frequency grid does not imply correspondingly fine physical resolution.

(h) Determine the number of positive-frequency bins normally displayed for a real signal.

Answer.

$$N/2 + 1 = 1024/2 + 1 = 513 \text{ bins}$$

(i) Describe the effect of replacing $L = 400$ by $L = 1600$ while keeping the hop duration proportional to the window length.

Answer.

$$\Delta f_{\text{new}} \approx 4 \cdot 16000/1600 = 40 \text{ Hz}$$

If the hop remains proportional to L , the time step also becomes four times larger. The FFT length must be at least the frame length, so $N=1024$ is no longer sufficient; a value such as 2048 is appropriate.

15. Spectrogram interpretation and analysis design

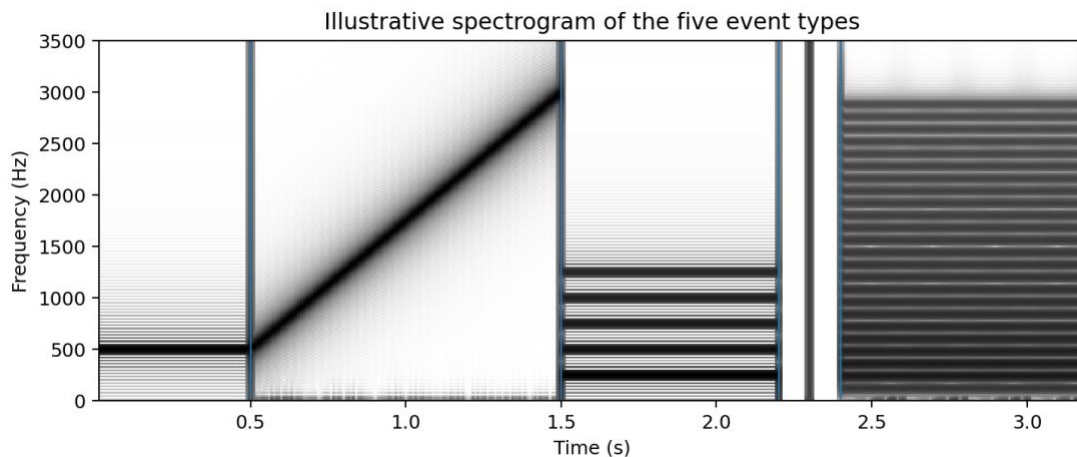
A recording contains five successive events:

- a 500-Hz pure tone lasting 0.5 s;
- a linear chirp rising from 500 Hz to 3000 Hz over 1 s;
- a periodic musical note with fundamental frequency 250 Hz and five harmonics;
- a short broadband impulse;
- a voiced speech segment.

For each event:

(a) Sketch or describe its expected spectrogram.

Answer.



Illustrative spectrogram of the five event types.

(b) Identify whether horizontal lines, diagonal ridges, harmonic stacks, broadband vertical structures, or formant regions should appear.

Answer. Each visual pattern is a direct time-frequency signature of the corresponding event type.

(c) Explain how a long analysis window would affect the display.

Answer. Long windows narrow spectral mainlobes and make harmonics easier to separate, but they average over a longer time interval. The impulse broadens in time and the chirp ridge becomes less sharply localized temporally.

(d) Explain how a short analysis window would affect the display.

Answer. The impulse becomes sharply localized in time, and fast changes are tracked better, but frequency resolution decreases.

(e) Choose an appropriate window length and overlap for distinguishing the events.

Answer. There is no single optimum for all five events. A moderate window around 25–32 ms and substantial overlap is a practical compromise, followed by task-specific adjustment.

(f) Explain whether linear magnitude or decibel magnitude would be more informative.

Answer. A dB scale compresses dynamic range so weak harmonics, sidelobes, and formant structure can be seen alongside strong components. Linear magnitude is useful when absolute amplitude ratios are the main concern.

(g) Describe one feature that could be misinterpreted as noise or as an additional component because of spectral leakage.

Answer. Leakage is especially problematic around strong stationary sinusoids and harmonics. A lower-sidelobe window helps distinguish true weak components from window artefacts.

Finally, write a short spectrum-analysis workflow covering sampling-rate verification, segment duration, window selection, FFT length, frequency-axis calibration, amplitude scaling, and validation with known signals.

Answer. A good analysis separates acquisition choices from display choices. First verify the sampling rate and absence of aliasing. Then choose segment duration and window according to the desired resolution. Use an FFT length at least as large as the frame, calibrate the frequency axis and amplitude scaling, and test the whole analysis on signals whose frequencies and amplitudes are known.